XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

Department of Electrical and Electronic Engineering

Xiamen University Malaysia

Quiz 1 (Q4)

EEE407 Electrical Energy Utilization

INSTRUCTIONS TO CANDIDATES

1. Use Microsoft Excel to solve this question, and upload the file to Moodle.
2. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you will be penalized accordingly.

Question 4 [CLO3-PLO2:C5] (30 marks)

The Matrix Corp started an energy project to reduce its energy usage. New LED lamps and air-conditioning and mechanical ventilation systems were installed in stages, which took place in the first two months of the year 2022. The energy consumption of the entire building prior and after the installation were recorded, as shown in Table 2. The monthly independent variables i.e., cooling degree days and number of employees, which could affect the energy consumption of the college building are also included.

a. With the appropriate baseyear, use regression method to determine the correlations between energy consumption and the two independent variables: CDD and number of employees/students. Plot the linear model and show the baseline equation and R^2 value for both independent variables. (10 marks)

Table 2: Energy consumption from January 2021 to February 2023.

Month-year	Cooling degree days (CDD)	Number of employees	Energy consumption (kWh)
Jan-21	287	54	15211
Feb-21	280	43	14280
Mar-21	295	43	16225
Apr-21	309	46	16068
May-21	328	52	17712
Jun-21	304	54	15808
Jul-21	300	55	15300
Aug-21	284	54	15336
Sep-21	270	56	13500
Oct-21	275	55	14025
Nov-21	279	53	15345
Dec-21	296	45	15096
Jan-22	299	32	5667
Feb-22	285	28	6025
Mar-22	301	49	10234
Apr-22	315	52	10080
May-22	334	58	11022
Jun-22	310	60	10850
Jul-22	306	61	9486
Aug-22	294	60	9114
Sep-22	286	62	9152
Oct-22	293	61	9083
Nov-22	291	59	9603
Dec-22	301	51	9331
Jan-23	320	38	10560
Feb-23	313	34	10642

b. Choose the suitable linear model, then determine the adjusted energy consumption (baseline energy) for each month and the monthly cumulative energy savings after the installation period. Show all data in a table and plot the energy profile (kWh vs. month) consists of all the stages: pre-retrofit, installation, post-retrofit, and baseline energy. (15 marks)

c. Estimate the simple payback period in years if the average energy cost is $0.50/kWh, and the total cost of the energy project is $250,000. (5 marks)